

EXPERIMENT 1 – PRIVATE CLOUD USING OPENSTACK

Aim

To create and configure a single-node private cloud environment using OpenStack MicroStack and to explore its major cloud services, including computing, networking, identity management, and storage. The experiment also involves creating and accessing virtual machine resources through the OpenStack dashboard.

Introduction to OpenStack

OpenStack is an open-source cloud computing platform used to build and manage cloud infrastructure. It is commonly used to provide **Infrastructure as a Service (IaaS)**.

With OpenStack, cloud administrators can manage a pool of physical resources such as:

- Compute resources
- Storage resources
- Network resources

Users can then access these resources and create virtual machines or other cloud resources when required.

OpenStack is commonly used for building **private and hybrid cloud environments**. In comparison, platforms such as AWS and Microsoft Azure mainly provide public cloud services.

What is MicroStack?

MicroStack is a lightweight OpenStack deployment designed to install and run OpenStack with minimal configuration.

It is especially useful for:

- Learning OpenStack concepts
- Testing cloud configurations
- Development purposes
- Running OpenStack on a single computer or virtual machine

In this experiment, MicroStack is installed on **Ubuntu 22.04 LTS** running inside VMware. The single-node setup allows the main OpenStack components to operate on one system.

Major OpenStack Services

OpenStack Service	Main Purpose
Nova	Creates and manages virtual machines
Neutron	Handles networking, IP addresses, and network connectivity
Keystone	Provides authentication and identity management
Glance	Stores and manages VM images
Horizon	Provides a web-based dashboard for OpenStack
Cinder	Provides persistent block storage for instances
Swift	Provides object-based storage

Software / Environment Used

- **Operating System:** Ubuntu 22.04 LTS (Jammy)
- **Cloud Platform:** OpenStack MicroStack
- **Virtualization Platform:** VMware
- **Deployment Type:** Single-node OpenStack
- **Interface:** Horizon Web Dashboard

Procedure

Step 1: Install MicroStack

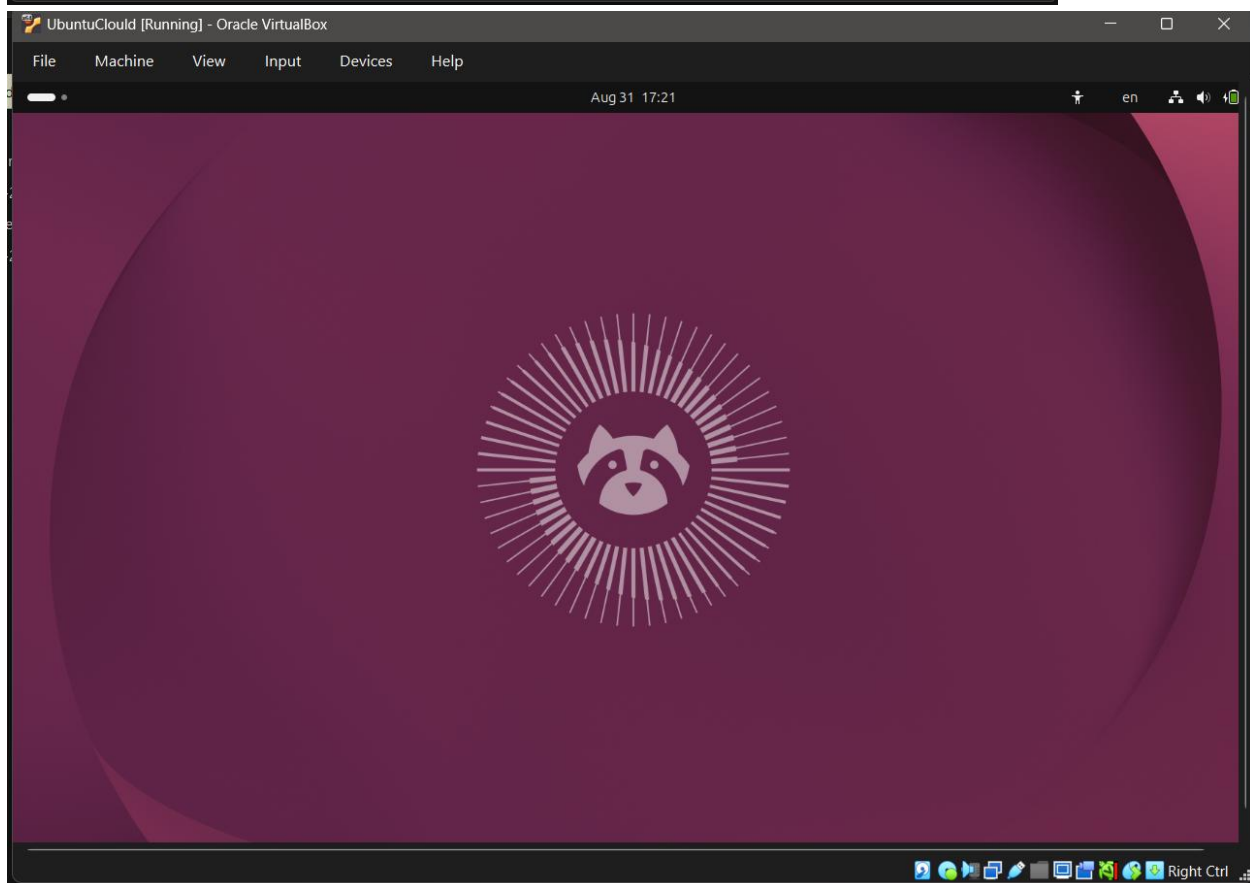
Open the Ubuntu terminal and install MicroStack using Snap:

```
sudo snap install microstack --beta
```

This downloads and installs the MicroStack package along with the required OpenStack components.

```
UbuntuCloud [Running] - Oracle VirtualBox
File Machine View Input Devices Help

[ OK ] Starting cloud-config.service - Cloud-init: Config Stage...
[ OK ] Started cups-browsed.service - Make remote CUPS printers available locally.
[ OK ] Starting systemd-user-sessions.service - Permit User Sessions...
[ OK ] Started ModemManager.service - Modem Manager
[ OK ] Finished systemd-user-sessions.service - Permit User Sessions.
[ OK ] Starting plymouth-quit-wait.service - Hold until boot process finishes up...
[ OK ] Started networkd-dispatcher.service - Dispatcher daemon for systemd-networkd.
[ OK ] Finished plymouth-quit-wait.service - Hold until boot process finishes up.
[ OK ] Starting setvtrgb.service - Set console scheme...
[ OK ] Finished setvtrgb.service - Set console scheme...
[ OK ] Finished cloud-config.service - Cloud-init: Config Stage.
[ OK ] Started snapd.service - Snap Daemon.
[ OK ] Starting snapd.hold.service - Holds Snappy daemon refresh...
[ OK ] Finished snapd.hold.service - Holds Snappy daemon refresh.
[ OK ] Started snap.desktop-security-center.hook.configure-910ffb59-1c8d-48ca-bd0a-a34281325bbc.scope.
[ OK ] Started snap.firefox.hook.install-fdb09973-8b34-462f-ace5-1550460c6b53.scope.
[ OK ] Started snap.firefox.hook.configure-8f72b384-f618-47b5-9abb-58982ec2365a.scope.
[ OK ] Started snap.firmware-updater.hook.configure-ba7e02eb-1546-4100-9014-2ba12a21a6cf.scope.
[ OK ] Started snap.mesa-2404.hook.install-0c0819ac-0ca6-4813-b1fb-c7adac65987f.scope.
[ OK ] Started snap.prompting-client.hook.configure-c37c372c-3f23-43fd-8171-7ebf8e0d1623.scope.
[ OK ] Started snap.snap-store.hook.configure-7d25b0d1-5806-4ffa-b82f-a8e2e7c490f6.scope.
[ OK ] Started snap.snapd-desktop-integration.hook.configure-3663a9f3-c90a-481f-aab6-1c5975c8ce0d.scope.
[ OK ] Started snap.thunderbird.hook.install-0384fcba-fa4a-41d6-9ece-9efd013bdc4e.scope.
[ OK ] Started snap.thunderbird.hook.configure-c977b964-9abf-49e0-b6e4-15f919680401.scope.
[ OK ] Started snap.ubuntu-desktop-bootstrap.subiquity-server.service - Service for snap application ubuntu-desktop-bootstrap.subiquity-server.
[ OK ] Started snap.ubuntu-desktop-bootstrap.hook.configure-32aef399-9536-4d00-9a4b-3dd5edafaf50.scope.
[ OK ] Stopping snap.ubuntu-desktop-bootstrap.subiquity-server.service - Service for snap application ubuntu-desktop-bootstrap.subiquity-server...
[ OK ] Stopped snap.ubuntu-desktop-bootstrap.subiquity-server.service - Service for snap application ubuntu-desktop-bootstrap.subiquity-server.
[ OK ] Starting systemd-timedated.service - Time & Date Service...
[ OK ] Started systemd-timedated.service - Time & Date Service.
[ OK ] Finished snapd.seeded.service - Wait until snapd is fully seeded.
[ OK ] Starting gdm.service - GNOME Display Manager...
[ OK ] Reached target multi-user.target - Multi-User System.
[ OK ] Starting cloud-final.service - Cloud-init: Final Stage...
[ OK ] Starting power-profiles-daemon.service - Power Profiles daemon...
[ OK ] Started casper-md5check.service - casper-md5Check Verify Live ISO checksums...
[ OK ] Started power-profiles-daemon.service - Power Profiles daemon.
[ OK ] Finished cloud-final.service - Cloud-init: Final Stage.
[ OK ] Reached target cloud-init.target - Cloud-init target.
[ OK ] Started gdm.service - GNOME Display Manager.
[ OK ] Reached target graphical.target - Graphical Interface.
[ OK ] Created slice user-1000.slice - User Slice of UID 1000.
[ OK ] Starting user-runtime-dir@1000.service - User Runtime Directory /run/user/1000...
[ OK ] Finished user-runtime-dir@1000.service - User Runtime Directory /run/user/1000.
[ OK ] Starting user@1000.service - User Manager for UID 1000...
[ OK ] Started user@1000.service - User Manager for UID 1000.
[ OK ] Started session-1.scope - Session 1 of User ubuntu.
[ OK ] Starting rtkit-daemon.service - RealtimeKit Scheduling Policy Service...
```

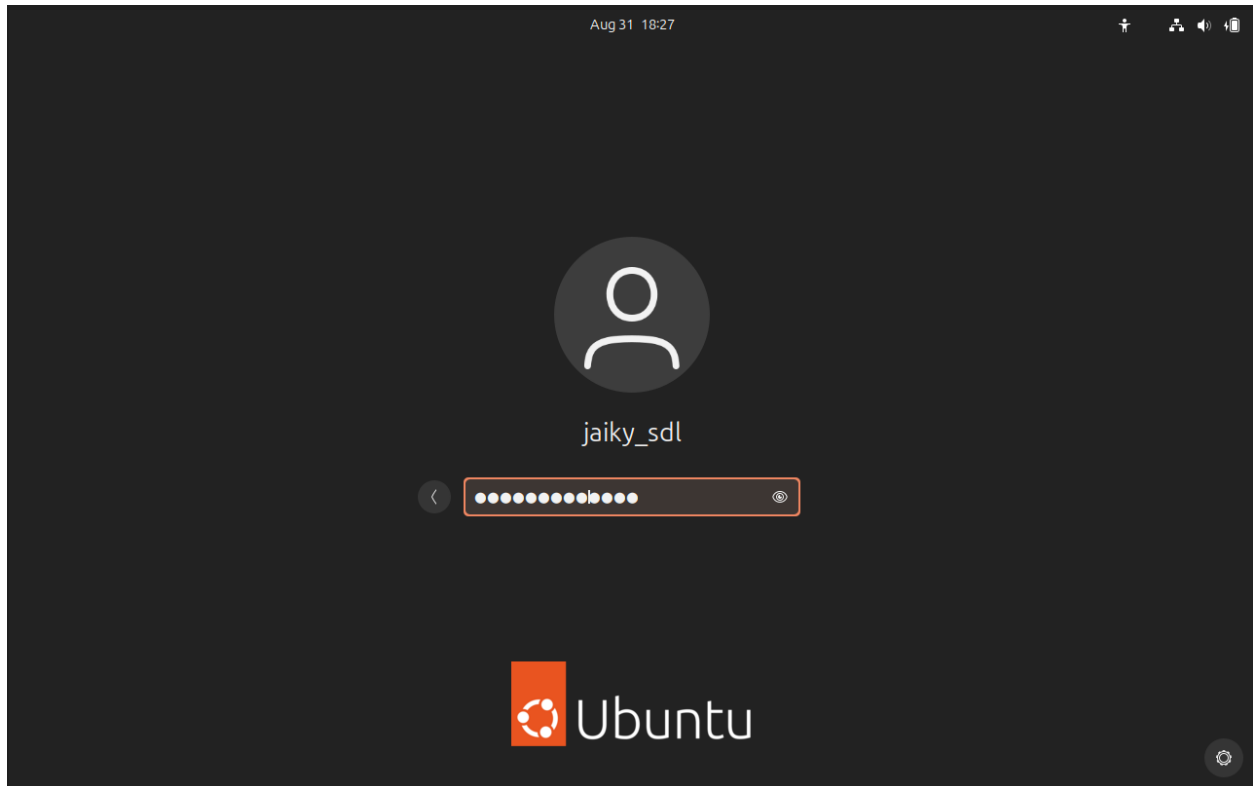


Step 2: Verify the Installation

To confirm that MicroStack has been installed successfully, execute:

```
snap list microstack
```

The installed MicroStack package and its version information will be displayed.



Step 3: Install Python Compatibility Package

Install the Python package required by the OpenStack environment:

```
sudo apt install python-is-python3
```

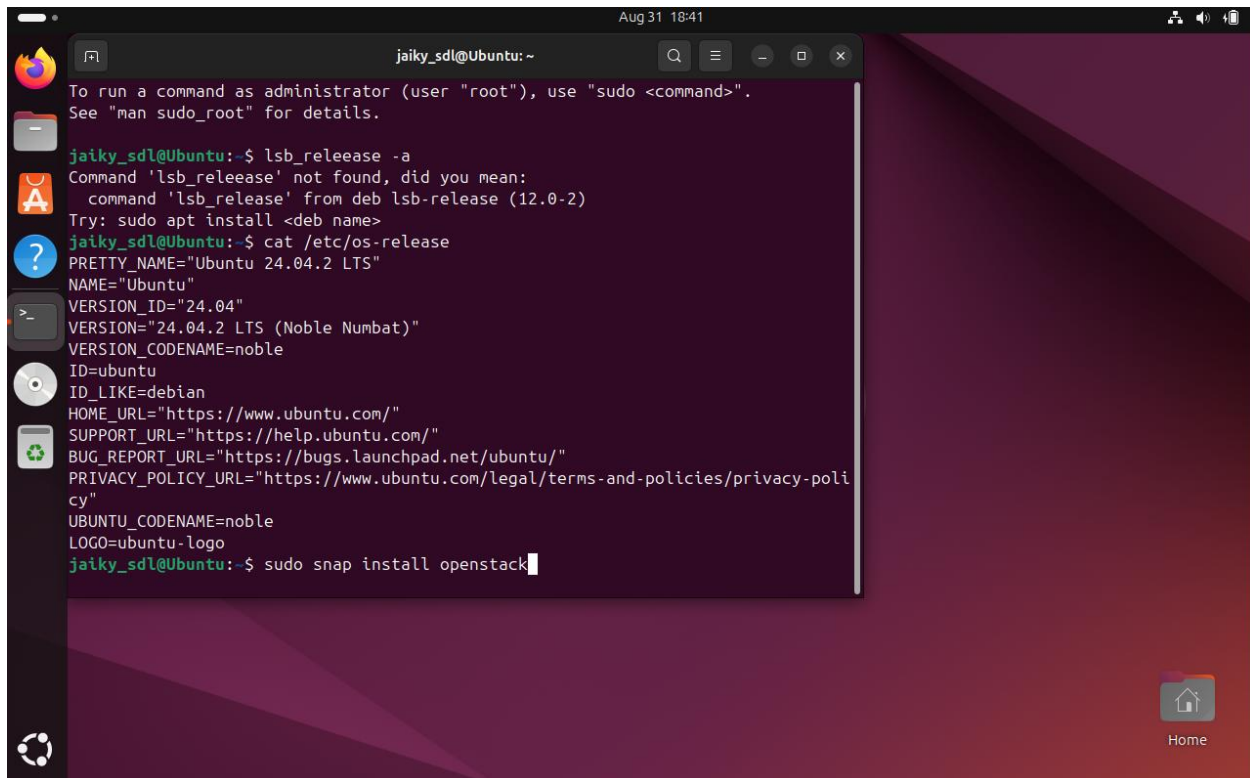
This ensures that the python command points to the installed Python 3 interpreter.

Step 4: Initialize MicroStack

Initialize the OpenStack environment automatically using:

```
sudo microstack init --auto --control
```

The command configures the required OpenStack services and prepares the system for use as a single-node cloud environment.



```
jaiky_sdl@Ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
jaiky_sdl@Ubuntu:~$ lsb_release -a  
Command 'lsb_release' not found, did you mean:  
  command 'lsb_release' from deb lsb-release (12.0-2)  
Try: sudo apt install <deb name>  
jaiky_sdl@Ubuntu:~$ cat /etc/os-release  
PRETTY_NAME="Ubuntu 24.04.2 LTS"  
NAME="Ubuntu"  
VERSION_ID="24.04"  
VERSION="24.04.2 LTS (Noble Numbat)"  
VERSION_CODENAME=noble  
ID=ubuntu  
ID_LIKE=debian  
HOME_URL="https://www.ubuntu.com/"  
SUPPORT_URL="https://help.ubuntu.com/"  
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"  
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"  
UBUNTU_CODENAME=noble  
LOGO=ubuntu-logo  
jaiky_sdl@Ubuntu:~$ sudo snap install openstack
```

Step 5: Access the OpenStack Dashboard

After the initialization is completed, open a web browser in the Ubuntu environment and enter the OpenStack dashboard address:

10.20.20.1

The **OpenStack Horizon** dashboard will appear.

Step 6: Login to Horizon

Use the following credentials to access the dashboard:

Username:

admin

To retrieve the automatically generated Keystone administrator password, run:

```
sudo snap get microstack config.credentials.keystone-password
```

Copy the displayed password and use it to log in to the Horizon dashboard.



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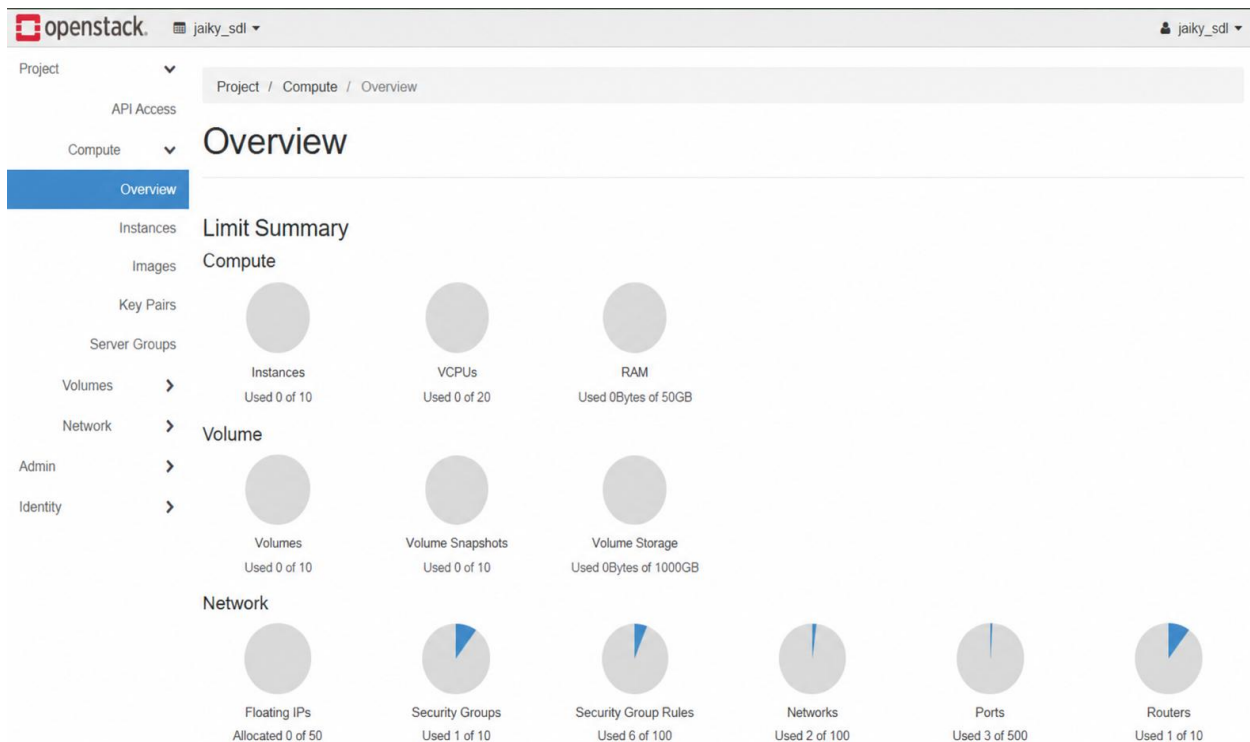
Log in

User Name

Password



Sign In



Conclusion

Thus, the basic setup of a private cloud using OpenStack MicroStack was successfully completed. The single-node deployment also demonstrated how OpenStack can be used for learning, testing, and development purposes.